Assessing spoken discourse in aphasia using multimodal artificial intelligence

Introduction: Narrative summarization of naturalistic stimuli such as movie clips is a functional task that approximates the demands of everyday communication, including integration of multimodal information and contextual cues over time. Main concept analysis (MCA) offers a criterion-based approach to quantifying narrative informativeness [1], yet labor-intensive manual inventory development and challenges in standardizing inventories have limited clinical implementation. Recent computational advances enable automated MCA [2,3], but prior work relies on text-based input and static stimuli, precluding assessment of the multimodal integration inherent to functional discourse. We employed a multimodal large language model (LLM) to generate MCA inventories from a set of movie clips We computed main concepts (MCs) and other discourse features in short narrative summaries of the clips obtained in persons with aphasia (PWA) and healthy controls, and tested whether the features discriminated between groups and predicted clinical severity of aphasia.

Methods: Fifty-four PWA (mean age 59.5±10.6 years; Western Aphasia Battery-Revised [WAB-R] Aphasia Quotient [AQ] 83.8±16.4) and 51 healthy controls (mean age 48.9±16.0 years) viewed and verbally summarized (limited to 1 minute) 8 English-language movie clips (140-348 seconds) from the Dynamic Affective Movie Clip Database [4]. Gemini 2.5 Pro was used to generate evidence-grounded MC inventories based on each clip's subtitles, audio, and video keyframes (up to 24 per clip, selected via shot-boundary detection), with prompt requiring supporting evidence for each concept and coverage spanning beginning, middle, and end of each clip's narrative. For the participants' narrative summaries, three MC features were computed relative to the inventory: distance to centroid (semantic deviation), coverage percentage (narrative completeness), and semantic coherence [5]. Traditional discourse features (n=17) were also computed using *dptools* [6], including lexical diversity (e.g., type-token ratio), lexical frequency (e.g., frequency range), productivity (e.g., word count), syntactic complexity (clauses per sentence), fluency (disfluencies, filled pauses), and part-of-speech ratios. Sentiment of discourse was computed using Gemma 3 (12B) [7]. Group discrimination was assessed using Cohen's *d* effect sizes (age-adjusted ANCOVA) and LASSO classification with leave-two-out cross-validation. Aphasia severity associations were examined via Pearson correlations with WAB-R AQ in the PWA subsample. All p-values were FDR-corrected for multiple comparisons.

Results: Main concept features showed the largest group differences and severity correlations. LASSO classification achieved excellent discrimination (AUC=0.917, accuracy 89.5%, sensitivity 85.2%, specificity 94.1%), with MC distance to centroid emerging as the strongest predictor (Figure 1). MC distance to centroid emerged as the strongest predictor for both group discrimination (d=1.46, p<.001) and severity correlation (r=−.78, p<.001), followed by disfluencies (d=1.07) for group effects and frequency range (r=.69) for severity prediction. Overall, PWA produced fewer unique main concepts (d=-1.81), exhibited more disfluencies (d=1.07), and showed reduced lexical diversity (d=-.84, all p<.001). Interestingly, within PWA, narrative sentiment was associated with aphasia severity (r=.63 with AQ, p<.001), with more severe impairment linked to more negatively rated retellings.

Discussion: MC features outperformed traditional microlinguistic metrics for both severity prediction and group classification, consistent with prior evidence that macrolinguistic informativeness measures are more sensitive to aphasia than microlinguistic indices [1]. This pattern likely reflects that successful narrative retelling requires integration across multiple language domains, e.g., lexical retrieval, morphosyntactic encoding, and pragmatic selection, such that MC-based measures capture cumulative impairment that isolated microstructural metrics miss. That semantic deviation from narrative centroids emerged as the single strongest predictor suggests global narrative disorganization, rather than any individual linguistic deficit, may best characterize discourse impairment in aphasia. The strong performance of LLM-based MC and sentiment features further demonstrates that automated multimodal analysis can extract clinically meaningful macrolinguistic information.

Figures and References

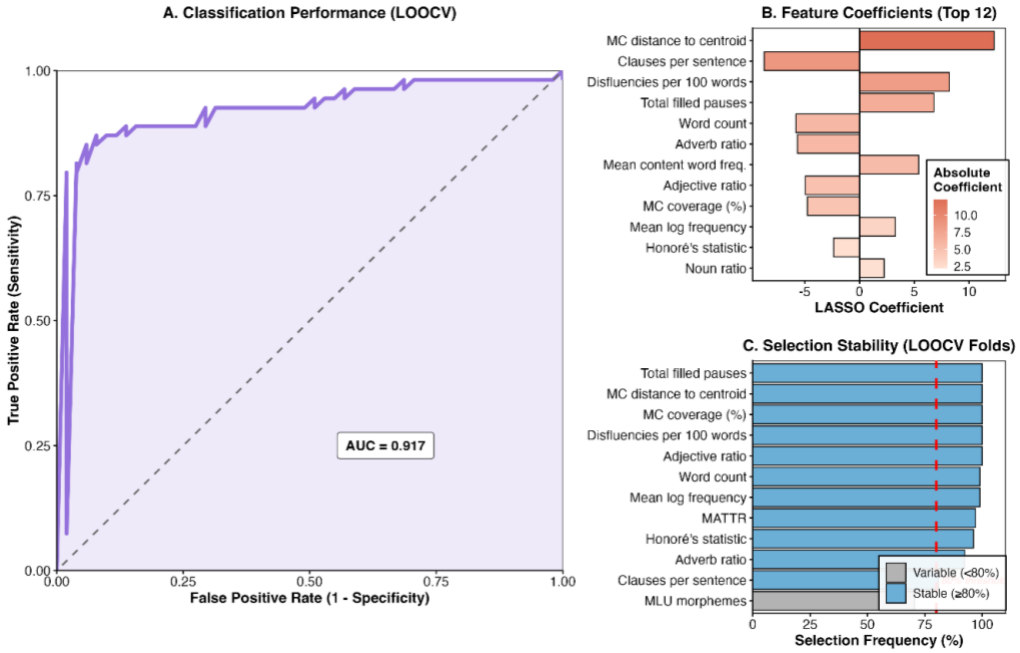

Figure 1. LASSO Classification Performance. (A) ROC curve from cross-validation. (B) Feature coefficients for top predictors with MC distance to centroid showing the largest coefficient. Bars colored by magnitude; positive (rightward) indicate PWA elevation, negative (leftward) indicate HC elevation. (C) Feature selection stability across cross-validation folds, showing percentage of iterations in which each feature received non-zero coefficient.

References

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